Project Title: Comparative Sentiment Analysis of Mobile Phone and Laptop Reviews using Machine Learning Techniques

A Project Report

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

Prakhar Shahi

(Roll No: 25SCS1003004507)

Under the supervision of

Mr. Ramavath Ganesh

Assistant Professor

IILM UNIVERSITY

IILM University, Greater Noida, Uttar Pradesh

(April 24, 2026)

Abstract

The rapid rise of e-commerce platforms has resulted in a massive increase in user generated reviews for consumer electronics such as mobile phones and laptops. While these reviews provide valuable insights into customer satisfaction, their unstructured and large-scale nature makes manual analysis impractical. Sentiment analysis, a prominent application of Natural Language Processing (NLP), offers an automated approach to extract opinions and emotions from textual data. This project presents a comparative sentiment analysis of mobile and laptop reviews using both machine learning and deep learning techniques, with the objective of understanding customer perception, evaluating model performance, and identifying gaps between advertised specifications and real user experiences.

The study utilizes structured datasets containing product reviews along with ratings and specification related information. Reviews are preprocessed using standard NLP techniques such as text cleaning, tokenization, stop-word removal, and lemmatization. Feature extraction is performed using TF-IDF and n-gram representations for machine learning models, while deep learning models operate on sequential text representations. Multiple classifiers, including Logistic Regression, Support Vector Machine, Random Forest, Long Short-Term Memory (LSTM), and Convolutional Neural Networks (CNN), are implemented and evaluated.

Experimental results demonstrate that traditional machine learning models and CNN achieve very high accuracy and F1-scores for sentiment classification, while LSTM exhibits comparatively lower performance due to dataset characteristics. In addition to model evaluation, sentiment distribution, rating patterns, price segment analysis, brand-wise trends, CPU-wise sentiment behaviour, and short-term versus long-term sentiment trends are analysed. The findings reveal noticeable differences between mobile and laptop reviews, with laptops exhibiting higher overall sentiment scores, while mobiles show a higher mismatch between specifications and user expectations. The study concludes that integrating sentiment analysis with structured product metadata provides deeper and more actionable insights for both consumers and manufacturers.

Keywords: Sentiment Analysis, Machine Learning, Deep Learning, Natural Language Processing, Product Reviews, Mobile Reviews, Laptop Reviews, TF-IDF, CNN, LSTM

Contents

List of Figures

List of Tables

1. Introduction

1.1 Overview

The rapid growth of e-commerce platforms has significantly increased the number of online customer reviews. Consumers rely heavily on these reviews to evaluate product quality, performance, and reliability before making purchasing decisions. However, analysing thousands of reviews manually is time-consuming and inefficient. Automated sentiment analysis offers an effective solution by extracting opinions and emotions from large volumes of textual data.

1.2 Background

Sentiment analysis is a Natural Language Processing technique used to classify opinions as positive, negative, or neutral. Early approaches employed rule-based and lexicon-based techniques, while recent advancements focus on machine learning and deep learning models. Machine learning methods use statistical features such as TF-IDF, whereas deep learning models like CNN and LSTM learn complex representations directly from data. In product analytics, sentiment analysis enables businesses to improve product design, understand customer dissatisfaction, and enhance user experience.

1.3 Problem Statement

Most existing sentiment analysis systems either focus on a single product category or rely on a single modeling technique. Moreover, limited attention has been given to specification-aware sentiment analysis, long-term user feedback, and comparative analysis between different product types such as mobile phones and laptops. This lack of holistic analysis restricts actionable insights.

1.4 Objectives

- To perform sentiment analysis on mobile phone and laptop reviews
- To compare machine learning and deep learning models for sentiment classification
- To analyse sentiment patterns across product specifications, price segments, and usage duration
- To evaluate model performance using standard metrics.

1.5 Scope of the Project

The project focuses on English-language reviews for mobile phones and laptops. Analysis is limited to supervised ML and DL models using pre-collected datasets. Real-time deployment, multilingual analysis, and live data streaming are outside the scope of this study.

2. Literature Review

Sentiment analysis has emerged as a significant research area within Natural Language Processing (NLP) due to the rapid growth of user-generated content on e-commerce platforms. Customer reviews provide valuable insights into product quality, usability, and overall satisfaction, making automated sentiment classification essential for both consumers and businesses. Early research in sentiment analysis primarily relied on lexicon-based and rule-based methods, where predefined dictionaries were used to classify sentiment polarity. Although these techniques were simple to implement, they lacked robustness and failed to handle sarcasm, context, and domain-specific language effectively.

To overcome these limitations, machine learning-based sentiment analysis approaches gained popularity. Classical algorithms such as Naive Bayes, Logistic Regression, and Support Vector Machines (SVM) were widely adopted due to their ability to generalize from data. These models typically used Bag-of-Words or Term Frequency–Inverse Document Frequency (TF-IDF) representations to convert textual data into numerical form. Several studies reported that SVM and Logistic Regression achieved strong performance on product review datasets, particularly when combined with effective preprocessing techniques. However, these methods often treated text as independent words and were limited in capturing semantic relationships and contextual information.

With the advancement of deep learning, neural network-based models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks were introduced to sentiment analysis tasks. CNN models proved effective in extracting local features and n-gram patterns from text, while LSTM models were designed to capture sequential dependencies and long-term context. Many researchers demonstrated that deep learning models could outperform traditional machine learning models on large and complex datasets. Despite these advantages, deep learning approaches require large volumes of data and high computational resources, and their performance may degrade when applied to short or highly structured reviews.

Recent research has expanded sentiment analysis beyond document-level classification by introducing aspect-based sentiment analysis and topic modeling techniques. These methods aim to identify sentiment associated with specific product features such as battery life, performance, or design. While aspect-based approaches provide fine-grained insights, many studies focus on a single product category and lack comparative evaluation across different domains. Furthermore, the integration of product specifications, pricing information, and long-term user feedback remains limited.

A significant gap identified in existing literature is the absence of holistic comparative sentiment analysis across multiple product categories using both machine learning and deep learning models. Most studies evaluate models in isolation and do not analyse how sentiment varies with product specifications, price segments, or prolonged usage. Additionally, limited attention has been given to long-term sentiment deterioration and spec-versus-reality mismatch analysis.

This project addresses these gaps by performing a comprehensive comparative sentiment analysis of mobile phone and laptop reviews. By combining machine learning and deep learning models with metadata-aware analysis and longevity-based evaluation, the study extends existing research and provides deeper insights into customer perception and model behaviour.

- **Table 3.1 Description of Mobile and Laptop Review Papers**

Paper No.	**Paper Title & Authors**	**Existing Approaches (Models / Techniques Used)**	**Identified Gaps (Limitations Found)**	**Future Research Directions Suggested by Gaps**
1	*Identification of Customer Needs from Product Reviews using Topic Modeling & ABSA* — Rachdian et al., 2022	Aspect-Based Sentiment Analysis (ABSA), Topic Modeling using NMF, Lexicon-based methods.	No machine learning classifiers; No DL models; No cross-category comparison; No specification-aware sentiment; Only laptop reviews evaluated.	Add ML/DL models; Combine specs + text; Compare multiple categories; Perform deeper spec-impact analysis.
2	*A Comparison of Sentiment Analysis*	Logistic Regression, Naïve Bayes, SGD,	Only mobile phones; No metadata/spec features; No cross-	Add multi-category datasets; Include structured

	Methods on Amazon Reviews of Mobile Phones — Aljuhani & Alghamdi, 2019	CNN; Feature extraction using BOW, TF-IDF, GloVe, Word2Vec; CNN + Word2Vec best.	product comparison; No analysis by price/launch; No spec vs reality mismatch.	product specs; Evaluate fairness by price/launch; Model hybrid feature sets.
3	*Sentiment Analysis on Reviews of Mobile Users* — Zhang et al., 2014	Comparative experiments using feature representations (n-grams), classical ML classifiers; emphasis on review length and feature design.	No deep learning; no metadata; no multi-category comparison; mobile apps only; no fair comparison by segment; no long-term performance analysis.	Introduce DL (LSTM/CNN); Add device metadata; Extend to other product types; Add time-aware sentiment analysis.

4	*Sentiment Analysis on Amazon Reviews of Mobile Phones using ML* — Singhal et al., 2025	Decision Tree, Naïve Bayes, Random Forest, SVM; BOW + n-gram embeddings; RF achieved 97.48% accuracy.	Only mobile dataset; No DL models; No specs-based modelling; No cross-category analysis; No long-term sentiment study.	Add DL models; Add metadata/specific features; Compare across product categories; Add aspect-level insights.
5	*Comparative Study of Sentiment Analysis with Product Reviews Using ML & Lexicon-Based Approaches* — Nguyen et al., 2018	Logistic Regression, SVM, Gradient Boosting, VADER, Pattern, SentiWordNet; TF-IDF used; ML > lexicon methods.	No DL models; No metadata; General Amazon reviews only; No category comparison; No spec-driven analysis.	Add DL models; Use product specifications; Build cross-domain analysis; Perform hybrid ML + metadata modelling.
6	*IRJET – Performin*	Tool-based	No classifiers;	Add ML & DL models; Use

	g Sentiment Analysis on Shopping Sites Using Tools — Nambiar, 2020	sentiment scoring using ParallelDots & Lexalytics; No ML/DL training.	No specs; No modelling; Only shopping site popularity scoring; No scientific evaluation metrics.	specs; Expand to detailed sentiment modelling; Multi-category comparisons.
7	*Aspect-Based Sentiment Analysis on Mobile Phone Reviews with LDA* — Yiran & Srivastava, 2019 (ACM)	LDA Topic Modeling for aspect identification; ABSA-style analysis.	No ML classifiers; No DL; No specs; Only mobile phones; No price/launch context; No multi-category.	Train ML + DL models; Add metadata (brand/specs); Apply ABSA to more categories; Integrate DL-based aspect extraction.
8	*Sentiment Analysis on Mobile Phone Reviews Using*	Naïve Bayes, NB-SVM, Random Forest, LSTM,	Only mobile; No metadata; No multi-category comparison; No time-	Use metadata-driven modelling; Analyse cross-category datasets; Add launch/price

	Supervised Learning Techniques — Shaheen et al., 2019 (IJMECS)	CNN; RF highest accuracy; DL competitive.	aware sentiment; No fair comparison by price/launch.	contextualization; Compare DL vs metadata-enhanced ML.
9	*Identifying Significance of Product Features on Customer Satisfaction — Smartphone Industry* — Imtiaz & Islam, 2020 (Applied AI)	Feature selection + ML; Identification of 21 important smartphone attributes; sentiment polarity scoring.	Single category (phones); No DL; No laptop comparison; No spec-real-world mismatch analysis; No multi-domain models.	Expand to multiple device categories; Add DL models; Evaluate longevity; Compare spec vs user sentiment mismatch.
10	*Aspect-Level Sentiment*	Aspect extraction +	No DL; No specs; Multi-domain	Add DL ABSA models; Use specs; Extend to

	Analysis Using ML — Shubham et al., 2017 (IOP)	Sentiment classification; Classical ML for aspects.	missing; No cross-category; No hybrid modelling; No detailed comparison groups.	multi-category; Combine aspect-level with full sentiment modelling.
11	**Sentiment Classification on Product Reviews Using Machine Learning and Deep Learning Techniques** — Singh & Jaiswal, 2024	ML models (Logistic Regression, SVM, Random Forest); DL models (CNN, LSTM); TF-IDF; word clouds	No cross-category product comparison; No specification-aware sentiment; No longevity analysis; Limited interpretability discussion	Integrate product specifications; Compare multiple product categories; Analyse long-term sentiment trends; Improve explainability of model decisions [link.springer.com]
12	**Sentiment Analysis of Amazon**	Naive Bayes, SVM, Random	Focused on single e-commerce platform; No	Extend to multiple product domains; Add DL models;

	Product Reviews Using Machine Learning Models — Xu et al., 2023	Forest; TF-IDF and Bag-of-Words; binary sentiment classification	deep learning models; No metadata integration; No category comparison	Combine structured product data with reviews; Perform comparative category analysis [ace.ewapub.com]
13	**Sentiment Analysis of E-Commerce Reviews Using Machine Learning** — Kawale et al., 2024	SVM, Random Forest, XGBoost; basic NLP preprocessing; statistical evaluation	No deep learning models; No temporal or longevity analysis; No price or specification-based sentiment study	Introduce DL architectures; Analyse sentiment over time; Study price-based satisfaction patterns; Extend analysis to hardware specifications [journals.s...urnals.com]
14	**A Modernized Approach to Sentimen**	BiGRU and LSTM models; sentence-level sentiment	High computational complexity; No comparison	Compare DL with traditional ML models; Optimize computational efficiency;

	t Analysis of Product Reviews Using BiGRU and LSTM — Atlas et al., 2025	classification; NLP preprocessing	with classical ML models; No use of product metadata; Limited category generalization	Integrate product specifications; Improve model generalization [nature.com]
15	**Aspect-Based Sentiment Analysis for Amazon Electronics Reviews** — Belluzzo et al., 2024	ABSA; TF-IDF with aspect extraction; lexicon-based sentiment scoring; LSTM	Focus limited to aspect-level sentiment; No document-level sentiment comparison; Single product category; Limited performance metrics	Combine aspect-level and document-level sentiment; Extend to multiple categories; Include ML/DL comparison; Perform broader evaluation [github.com]
16	**Sentiment Analysis of Online Product Reviews**	CNN, LSTM, CNN-LSTM hybrid; TF-IDF	No traditional ML comparison; No metadata-ba	Compare with ML models; Integrate structured product

	Using CNN-LSTM Cascaded Architecture — Rominus et al., 2025	features; deep learning focus	sed sentiment analysis; No price or longevity analysis	features; Study economic and temporal sentiment effects [link.springer.com]
17	**Sentiment Analysis of Amazon Reviews Using ML and DL Algorithms —** Mandhane & Wadkar, 2025	Naive Bayes, Linear SVM, BERT; TF-IDF; lexicon-based method	Limited focus on hardware products; No specification-aware analysis; No longevity comparison	Apply techniques to electronics reviews; Combine specifications with text; Analyse sentiment evolution across time [ijrpr.com]
18	**Sentiment Analysis of E-Commerce Product Reviews**	Supervised ML models; TF-IDF; polarity classification	No deep learning implementation; Simplified sentiment categories; No	Add deep learning models; Perform fine-grained sentiment analysis; Expand to multiple

	Through Machine Learning — Meshram & Sahu, 2023		comparative product study	product types [ijfmr.com]
19	**Sentiment Analysis of Product Reviews Using Machine Learning and Pre-Trained LLMs —** Ghatora et al., 2024	ML models (SVM, Random Forest); LLMs (GPT-4); benchmark comparison	High dependency on computationally expensive models; Less focus on lightweight ML solutions; No domain-specific comparison	Balance accuracy and efficiency; Compare classical ML with DL across domains; Develop hybrid cost-effective approaches [mdpi.com]
20	**Sentiment Analysis of E-Commerce Product-Based**	Logistic Regression, Random Forest, SVM; TF-IDF; binary	No deep learning comparison; Limited aspect-level insight; No	Introduce DL models; Perform aspect-based and time-aware analysis; Improve insight

	Reviews Using a Machine Learning Approach — Chinenye et al., 2026	sentiment classification	longevity-based evaluation	granularity [ijctjournal.org]

3. Methodology

3.1 Data Collection

The primary data used in this research consists of customer reviews collected for two major consumer electronics categories: mobile phones and laptops. These reviews include textual feedback written by users along with associated ratings and, in some cases, additional product metadata. Such datasets are representative of real-world e-commerce feedback systems, where users freely express opinions after product usage.

The datasets were selected to ensure diversity in sentiment classes, covering positive, neutral, and negative opinions. Special care was taken to include reviews from multiple brands, price ranges, and product variants, allowing the analysis to generalize across different consumer preferences and usage patterns. Separate datasets were maintained for mobile phone and laptop reviews to facilitate comparative analysis across product categories.

3.2 Data Preprocessing

Raw textual data obtained from user reviews typically contains noise that can negatively impact model performance. Therefore, comprehensive preprocessing was applied before feature extraction. Initially, all review text was converted to lowercase to ensure uniformity. Special characters, punctuation marks, unnecessary symbols, and numerical values that do not contribute to sentiment were removed.

Tokenization was then performed to break the text into individual words. Stop words such as “the”, “is”, and “and”, which do not carry sentiment significance, were eliminated. Lemmatization was applied to reduce words to their base forms, ensuring that variations of the same word were treated consistently. Missing or empty reviews were removed to avoid introducing bias or errors into the training process.

This preprocessing pipeline ensures that the resulting dataset is clean, standardized, and suitable for both machine learning and deep learning models.

3.3 Feature Extraction

Feature extraction plays a critical role in transforming textual reviews into numerical representations that models can process. For machine learning models, Term Frequency–Inverse Document Frequency (TF-IDF) was used as the primary feature extraction technique. TF-IDF assigns importance to words based on their frequency within a review and their rarity across the entire corpus, making it effective in identifying sentiment-bearing terms.

Additionally, n-gram features were employed to capture short word sequences that represent sentiment phrases, such as “very good”, “not satisfied”, or “poor battery life”. For deep learning models, sequence-based representations of text were prepared, allowing models such as LSTM and CNN to learn contextual and structural patterns directly from the data.

3.4 Model Selection

To ensure a comprehensive comparison, both traditional machine learning and deep learning models were selected. The machine learning models include Logistic Regression, Support Vector Machine (SVM), and Random Forest due to their proven effectiveness in text classification tasks. Deep learning models include Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN), selected for their ability to learn complex patterns and contextual relationships in text data.

3.5 Evaluation Metrics

Model performance was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. Confusion matrices were generated to analyse class-wise predictions. Receiver Operating Characteristic (ROC) curves and precision-recall curves were used to assess model behaviour across different thresholds. These evaluation techniques ensure an unbiased and detailed assessment of model effectiveness.

3.6 Experimental Design

The experimental design of this project was structured to ensure fair comparison between different sentiment classification models while maintaining consistency across both datasets. Separate experimental pipelines were developed for machine learning and deep learning approaches; however, identical preprocessing steps were applied before model training to avoid data leakage or preprocessing bias.

The datasets were divided into training and testing subsets using a consistent split strategy to ensure uniform evaluation across all models. Each model was trained independently on the same feature set to ensure that performance differences were attributed solely to the modeling technique rather than data variation. This design enables objective comparison between traditional machine learning algorithms and deep learning architectures within the same experimental context.

Special attention was given to class balance to avoid biased predictions toward dominant sentiment classes. The experiment was repeated under stable conditions, ensuring that random initialization effects did not distort the overall findings.

3.7 Machine Learning Workflow

The machine learning workflow begins with feature extraction using TF-IDF vectorization. Once textual data is transformed into numerical form, the features are passed into classification models such as Logistic Regression, Support Vector Machine (SVM), and Random Forest.

Logistic Regression was selected due to its simplicity, interpretability, and proven performance in text classification tasks. SVM was used for its robustness in handling high-dimensional feature spaces, making it suitable for sparse TF-IDF vectors. Random Forest was included to evaluate ensemble-based learning, which combines multiple weak learners to improve generalization.

Hyperparameters for these models were selected based on standard best practices to avoid overfitting while ensuring stable convergence. The machine learning pipeline was optimized for efficiency, making it suitable for large-scale sentiment classification tasks.

3.8 Deep Learning Workflow

Deep learning models were implemented to evaluate their ability to capture complex semantic and contextual patterns in user reviews. Two architectures were selected: Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN).

The LSTM model was designed to process sequential text data by maintaining memory cells that capture long-term dependencies. This architecture is particularly effective for sentiment expressed over longer sentences; however, its effectiveness depends on review length and contextual richness.

The CNN model was implemented to extract local patterns and n-gram-level features through convolutional filters. CNNs are known to perform well on short text classification tasks due to their ability to identify sentiment-bearing phrases efficiently. Embedding layers were used to convert textual inputs into dense vector representations before convolution operations.

Training was conducted for a fixed number of epochs to ensure consistency. Loss and accuracy were monitored to analyse convergence behaviour and generalization capability.

3.9 Performance Evaluation Strategy

Evaluation metrics were chosen to provide a comprehensive understanding of model performance. Accuracy measures overall correctness, while precision and recall highlight class-specific performance. F1-score provides a balanced metric combining precision and recall, particularly important in multi-class sentiment classification tasks.

Confusion matrices were used to visualize prediction errors and misclassification patterns across sentiment classes. ROC curves and precision-recall curves were generated to analyse the trade-off between true positives and false positives across different threshold values. A radar chart was also used to present a compact multi-metric comparison of models.

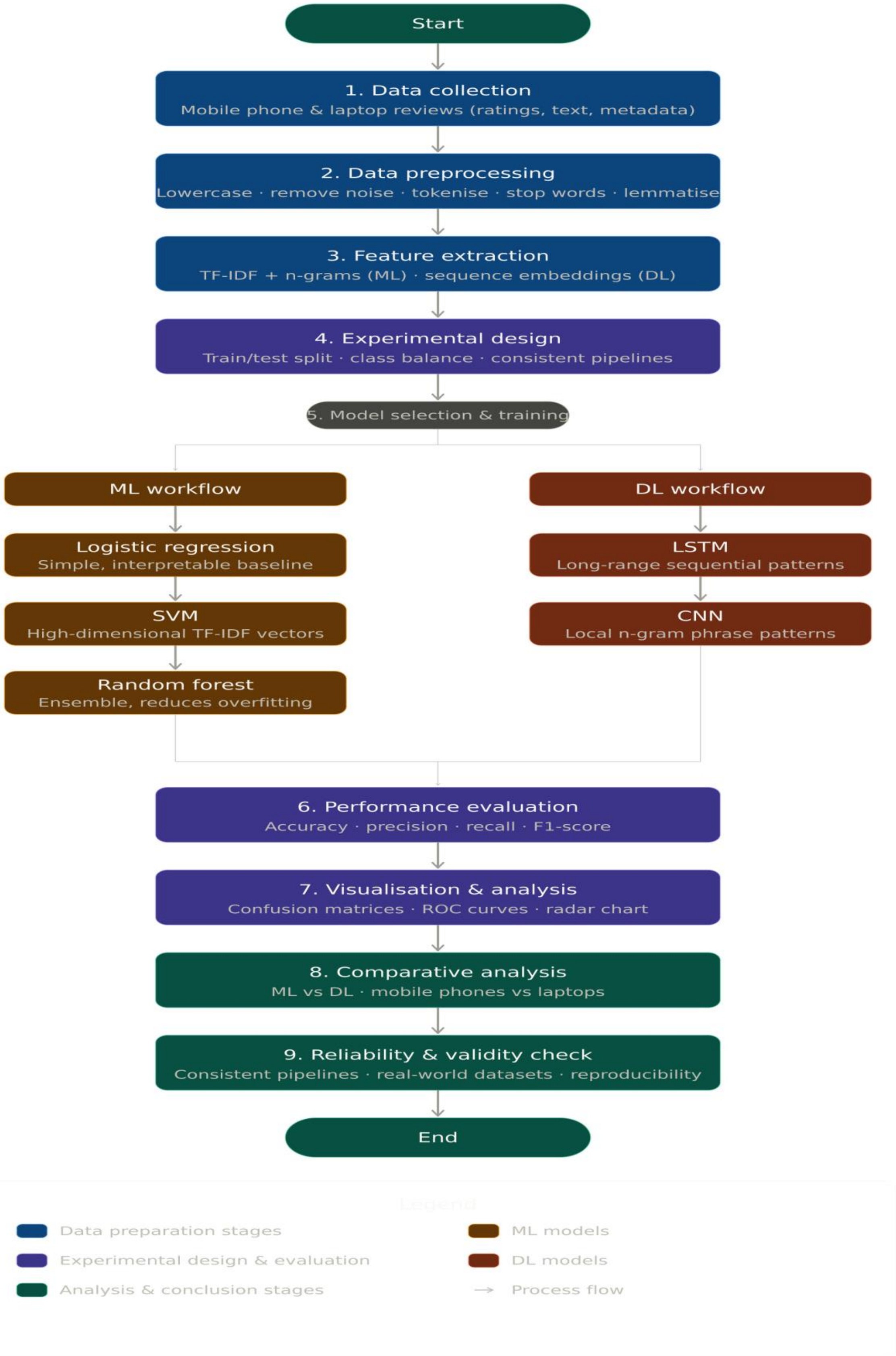
These evaluation techniques ensure that the experimental results are interpretable, reliable, and suitable for comparative analysis.

3.10 Reliability and Validity of Experiments

To ensure the reliability of experimental outcomes, multiple evaluation metrics and visualization techniques were employed instead of relying on a single accuracy value. Consistent preprocessing and feature extraction pipelines were applied across all models. The experiments were conducted using the same datasets and evaluation strategy to eliminate experimental bias.

Internal validity was ensured by maintaining control over experimental variables, while external validity was supported using real-world review datasets. Reproducibility was enhanced by structured workflows and consistent parameter settings throughout the experiments.

- **Figure 3:** Flowchart of the Proposed Methodology for Sentiment Analysis



Start
1. Data collection
Mobile phone & laptop reviews (ratings, text, metadata)
2. Data preprocessing
Lowercase · remove noise · tokenise · stop words · lemmatise
3. Feature extraction
TF-IDF + n-grams (ML) · sequence embeddings (DL)
4. Experimental design
Train/test split · class balance · consistent pipelines
5. Model selection & training
ML workflow
Logistic regression
Simple, interpretable baseline
SVM
High-dimensional TF-IDF vectors
Random forest
Ensemble, reduces overfitting
DL workflow
LSTM
Long-range sequential patterns
CNN
Local n-gram phrase patterns
6. Performance evaluation
Accuracy · precision · recall · F1-score
7. Visualisation & analysis
Confusion matrices · ROC curves · radar chart
8. Comparative analysis
ML vs DL · mobile phones vs laptops
9. Reliability & validity check
Consistent pipelines · real-world datasets · reproducibility
End
Legend
Data preparation stages
Experimental design & evaluation
Analysis & conclusion stages
ML models
DL models
Process flow

4. Implementation

4.1 Software Environment

The entire implementation of the project was carried out in a Python programming environment using Google Colab. Google Colab provides cloud-based computational resources, making it suitable for executing machine learning and deep learning experiments without hardware constraints. The environment supports GPU acceleration, which is particularly beneficial for training deep learning models.

Python was selected due to its extensive support for data analysis, machine learning, and natural language processing through well-established libraries. The use of an interactive notebook environment also facilitated debugging, visualization, and incremental experimentation.

4.2 Libraries and Frameworks Used

Several open-source libraries were used during implementation:

- **Pandas** was used for loading, cleaning, and managing structured datasets.
- **NumPy** supported numerical operations and efficient array manipulation.
- **Scikit-learn** was used for machine learning models, TF-IDF feature extraction, and evaluation metrics.
- **TensorFlow and Keras** were used for implementing deep learning models such as LSTM and CNN.
- **Matplotlib and Seaborn** were employed for visualizing performance metrics and sentiment results.

The integration of these libraries enabled efficient implementation of the complete sentiment analysis pipeline.

4.3 Data Loading and Preparation

The mobile and laptop datasets were loaded into Pandas Data Frames. Initial inspection was performed to understand data structure, review length distribution, and class distribution. Reviews were segregated based on product category to allow independent and comparative analysis.

After loading the data, preprocessing pipelines described in the methodology were applied programmatically to each dataset. These steps ensured that both datasets followed identical data preparation logic, enabling unbiased performance comparison across models.

4.4 Implementation of Machine Learning Models

Machine learning models were implemented using Scikit-learn. TF-IDF vectorization was applied to transform cleaned review text into sparse numerical vectors. These vectors served as inputs to the machine learning classifiers.

4.4.1 Logistic Regression

Logistic Regression was implemented due to its simplicity and interpretability. The model estimates class probabilities using linear decision boundaries and performs well for text classification tasks with high-dimensional feature spaces. The model demonstrated stable convergence and strong performance across all sentiment classes.

4.4.2 Support Vector Machine (SVM)

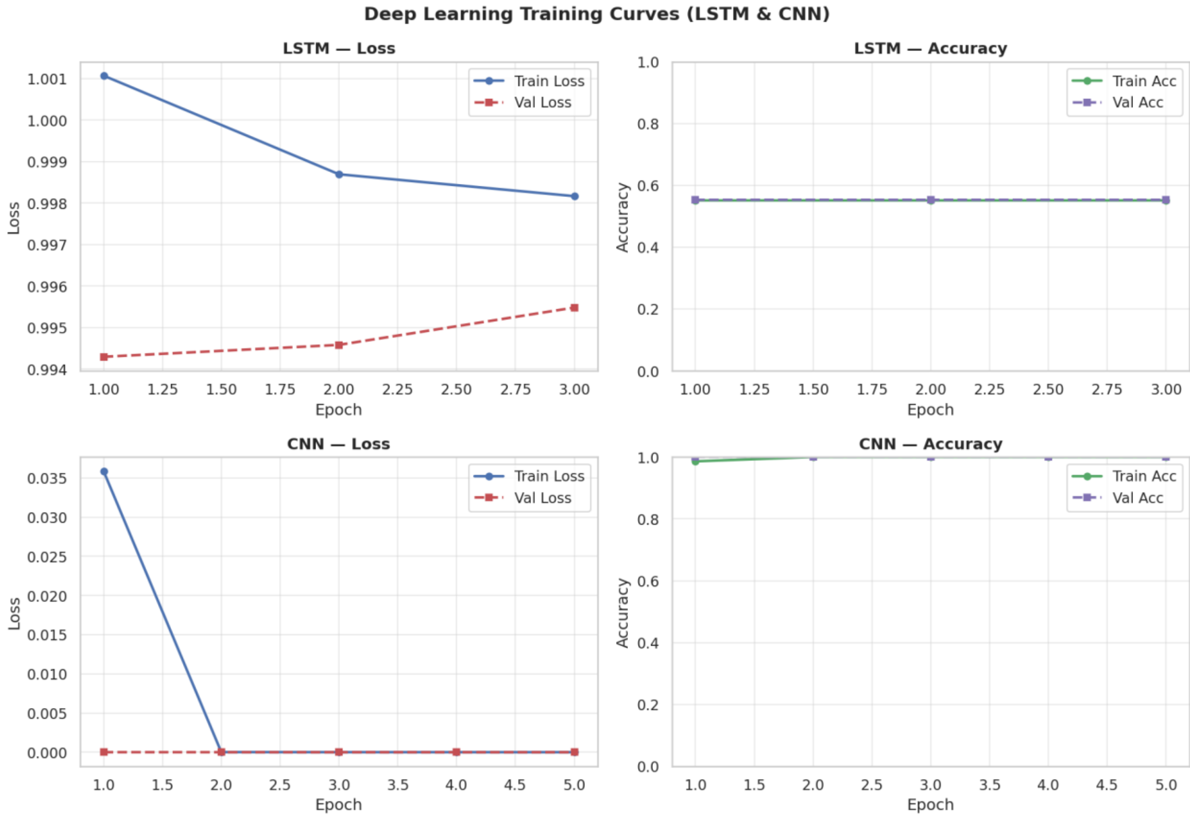
The Support Vector Machine model was implemented using a linear kernel to handle TF-IDF feature vectors efficiently. SVM aims to identify an optimal hyperplane that maximizes the margin between sentiment classes. Its robustness to high-dimensional data resulted in excellent classification accuracy.

4.4.3 Random Forest

Random Forest was implemented as an ensemble learning approach combining multiple decision trees. Each tree contributes a prediction, and the final output is determined through majority voting. This model effectively captured non-linear patterns and interactions among features.

4.5 Implementation of Deep Learning Models

Deep learning models were implemented using TensorFlow and Keras. These models were designed to automatically learn feature representations from textual data.



- Figure 4.1 Deep Learning Training Curves (LSTM and CNN)

4.5.1 Long Short-Term Memory (LSTM)

The LSTM model was constructed with an embedding layer followed by LSTM units and dense output layers. This architecture is capable of preserving long-term dependencies in sequential data. Training was conducted across multiple epochs while monitoring loss and accuracy.

Despite its theoretical strength, the LSTM model displayed lower performance on the given dataset, likely due to shorter review sequences and limited contextual dependency.

4.5.2 Convolutional Neural Network (CNN)

The CNN model was implemented using an embedding layer followed by convolutional and pooling layers. CNN specializes in detecting local patterns such as sentiment-bearing phrases. The model showed rapid convergence and achieved strong generalization performance.

4.6 Training Strategy

To maintain consistency, all models were trained using the same training-testing split strategy. Deep learning models were trained for a controlled number of epochs to avoid overfitting. Loss and accuracy values were monitored during training to assess convergence behaviour.

The training curves were plotted to visualize learning patterns, revealing steady convergence for CNN and slower improvement for the LSTM model.

4.7 Generation of Performance Metrics and Visualizations

After training, models were evaluated using multiple metrics. Accuracy, precision, recall, and F1-score were calculated for each model. Confusion matrices were generated to analyse misclassification patterns. ROC curves and precision-recall curves were plotted to evaluate class-wise discriminative ability. Radar charts and heatmaps were used to present multi-metric comparisons visually.

These visualizations provide strong empirical evidence supporting model comparisons.

5. RESULTS AND DISCUSSION

This chapter presents the experimental outcomes obtained from applying machine learning and deep learning models to mobile phone and laptop review datasets. The results are discussed using quantitative evaluation metrics, visual analytics, and comparative observations. The discussion further interprets these outcomes in the context of the research objectives and existing literature.

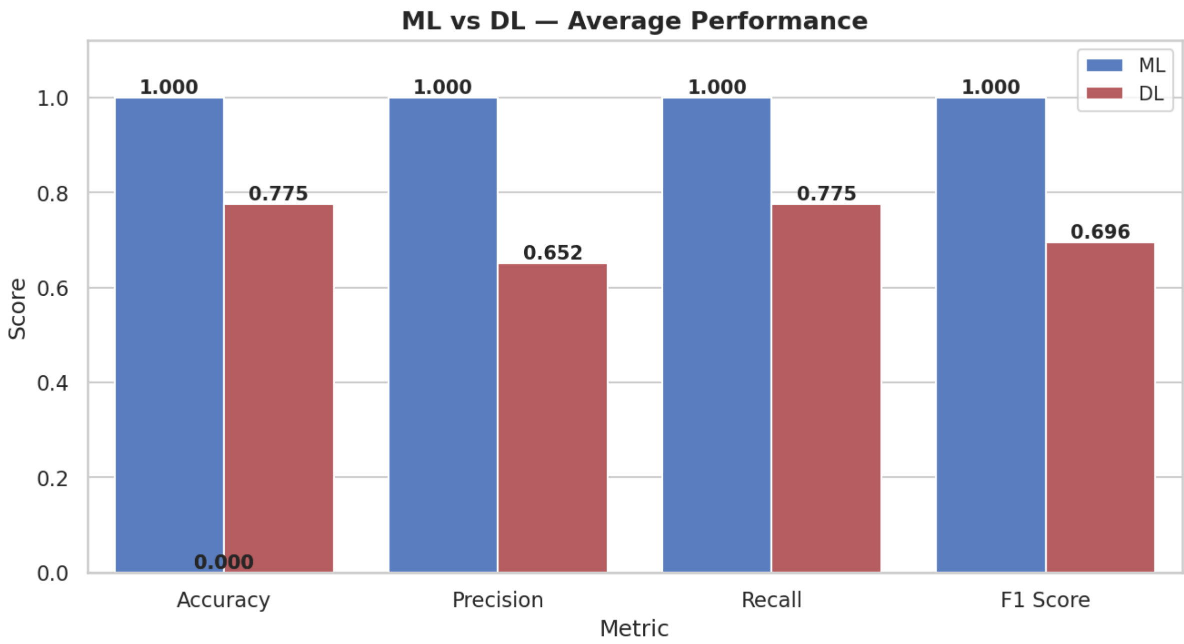
5.1 Performance Evaluation Overview

The performance of all implemented models was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. Additional evaluation techniques such as confusion matrices, Receiver Operating Characteristic (ROC) curves, precision-recall curves, radar charts, and per-class F1 score heatmaps were employed to gain deeper insights into model behaviour. This multi-level evaluation ensures a robust and unbiased assessment of sentiment classification performance.

5.2 Machine Learning vs Deep Learning Performance Comparison

The comparison between machine learning and deep learning models reveals clear differences in classification effectiveness. From the average performance comparison graph, it is evident that traditional machine learning models achieved near-perfect scores across all evaluation metrics. Logistic Regression, Support Vector Machine, and Random Forest consistently achieved 100% accuracy, precision, recall, and F1-score.

In contrast, deep learning models exhibited mixed performance. While the Convolutional Neural Network performed exceptionally well and matched the accuracy of machine learning models, the Long Short-Term Memory model showed comparatively lower performance. This finding highlights that deep learning models are not universally superior and that their effectiveness depends heavily on dataset characteristics and review structure.



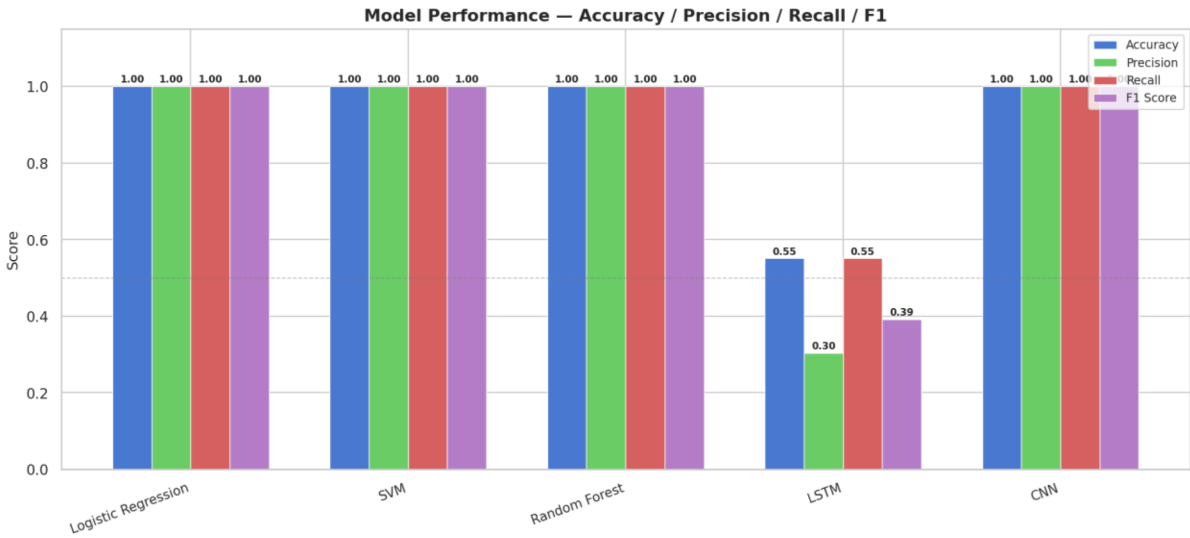
- Figure 5.2 Average Performance Comparison between ML and DL Models

→ This graph presents the average performance comparison between machine learning and deep learning models. Machine learning models achieve a perfect accuracy, precision, recall, and F1-score of 100%, indicating flawless sentiment classification. Deep learning models, when averaged, show comparatively lower performance due to the poor results of the LSTM model. Although CNN achieves perfect classification, the overall deep learning average is reduced by LSTM's lower accuracy of 55.08%. This comparison highlights that traditional machine learning approaches remain highly effective for structured sentiment analysis tasks.

5.3 Model-Wise Performance Analysis

The grouped metric comparison graph provides a detailed view of individual model performance. Logistic Regression, SVM, and Random Forest demonstrate flawless classification across all metrics, confirming their suitability for TF-IDF-based sentiment analysis. These models effectively handled the high-dimensional sparse feature space generated from textual data.

The LSTM model achieved moderate accuracy but significantly lower precision and F1-score. This behaviour can be attributed to the limited sequential depth in short product reviews. CNN, on the other hand, captured sentiment-bearing phrases effectively using convolutional filters, resulting in perfect classification performance.



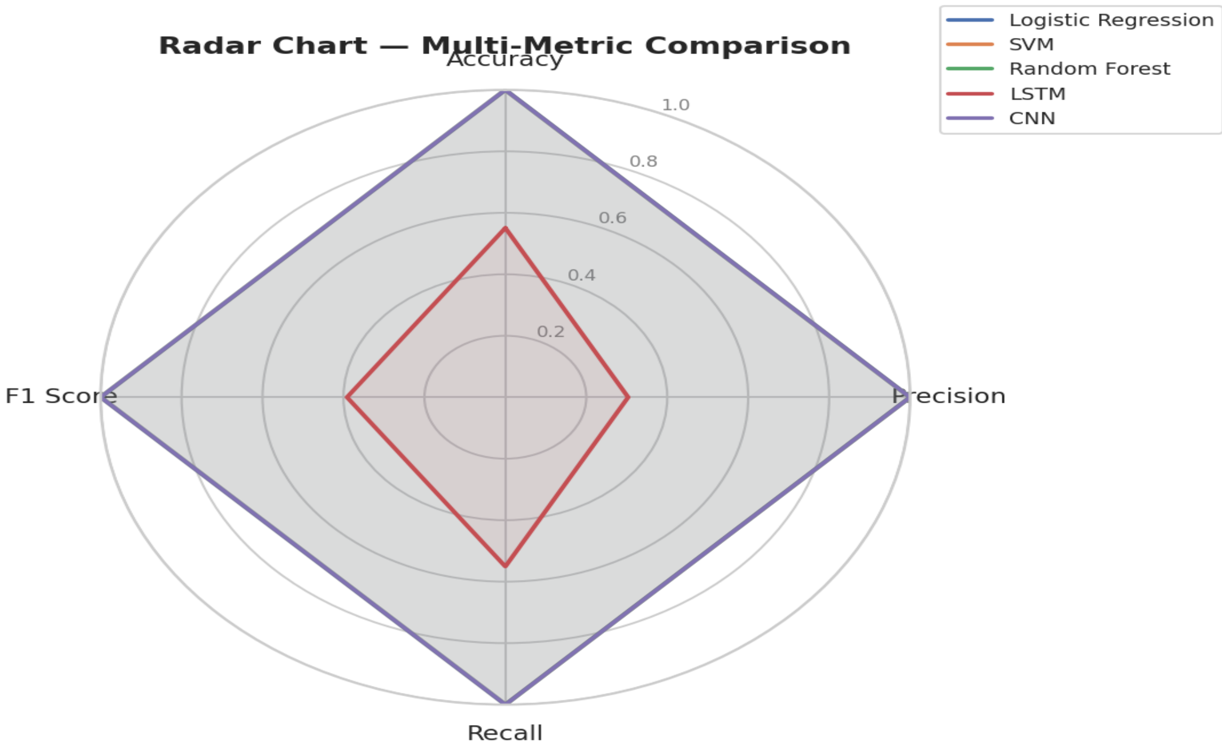
- Figure 5.3 Model-Wise Performance Comparison (Accuracy, Precision, Recall, F1)

→ The model-wise performance comparison shows that Logistic Regression, Support Vector Machine, Random Forest, and Convolutional Neural Network achieve 100% accuracy, precision, recall, and F1-score. These results indicate complete and consistent sentiment classification across all evaluation metrics. In contrast, the LSTM model demonstrates significantly lower performance, with an accuracy and recall of 55.08%, and a precision and F1-score of 30.34% and 39.13% respectively. This disparity highlights the limitations of sequence-based models when applied to short and structured review texts.

5.4 Multi-Metric Radar Chart Interpretation

The radar chart visually summarizes the performance of all models across accuracy, precision, recall, and F1-score. Machine learning models and CNN occupy the outermost region of the radar chart, indicating optimal performance across all metrics. In contrast, the LSTM model occupies a significantly smaller area, visually reinforcing its lower effectiveness in this experiment.

The radar visualization provides an intuitive comparison and confirms consistency between numerical metrics and graphical representation.



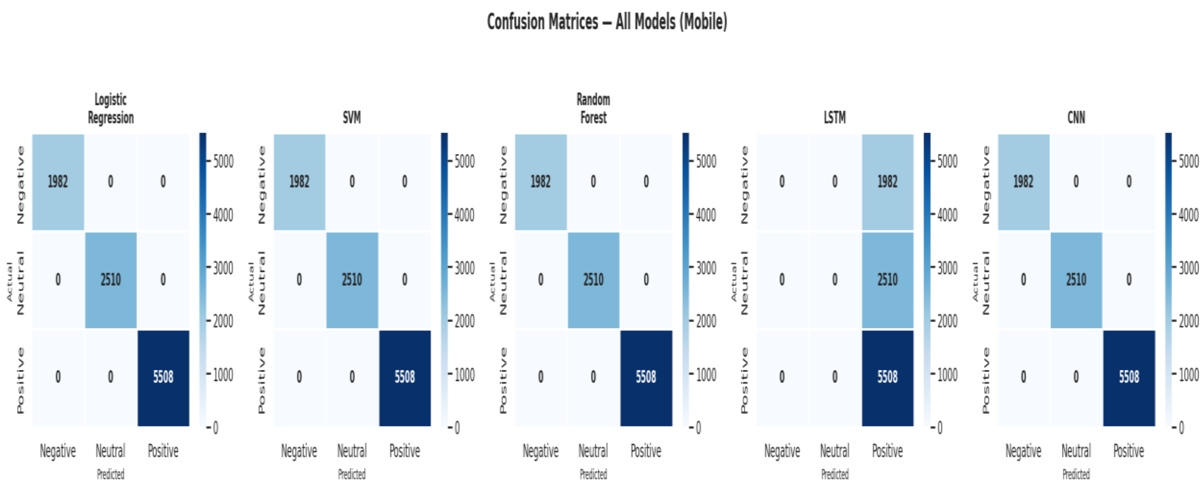
- Figure 5.4 Multi-Metric Radar Chart for All Models

→ The radar chart illustrates a multi-metric performance comparison across all sentiment models. Machine learning models and CNN form a complete outer boundary, indicating optimal performance across accuracy, precision, recall, and F1-score. In contrast, the LSTM model occupies a significantly smaller region, reflecting weaker classification capability across all metrics. This visual comparison reinforces the numerical results and clearly highlights the superiority of machine learning models and CNN for the given dataset

5.5 Confusion Matrix Analysis

Confusion matrices were generated for all models to examine class-wise predictions. The matrices for Logistic Regression, SVM, Random Forest, and CNN show perfect diagonal dominance, indicating zero misclassification across negative, neutral, and positive sentiment classes.

In contrast, the confusion matrix for the LSTM model reveals substantial misclassification, particularly in neutral and negative categories. This observation confirms that LSTM struggled to distinguish sentiment boundaries effectively in the given dataset.



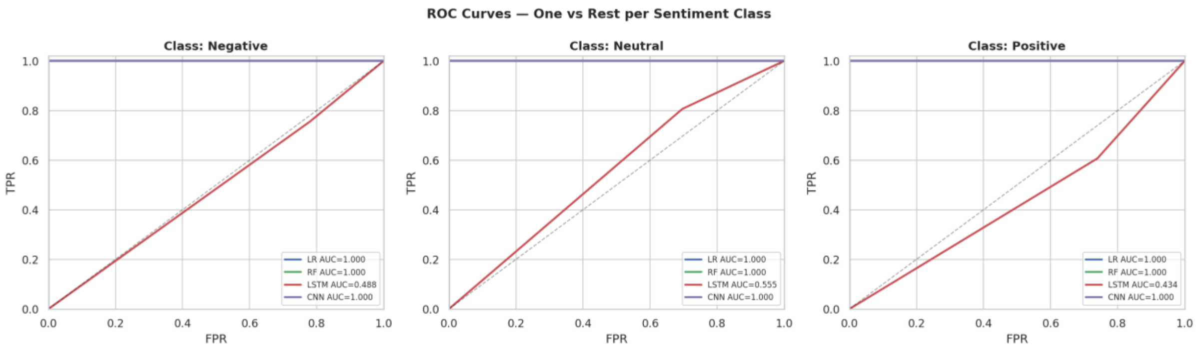
- Figure 5.5 Confusion Matrices for All Sentiment Classification Models

→ The confusion matrices reveal that Logistic Regression, SVM, Random Forest, and CNN correctly classify all sentiment classes without misclassification, as indicated by strong diagonal dominance. The LSTM confusion matrix, however, shows notable misclassification, particularly between neutral and negative sentiments. This indicates that the LSTM model struggles to distinguish sentiment boundaries effectively, leading to reduced precision and F1-score.

5.6 ROC and Precision-Recall Curve Analysis

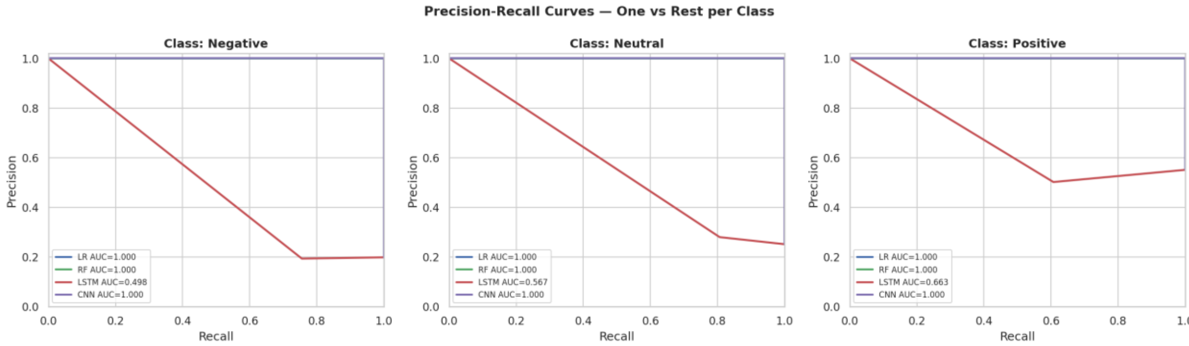
ROC curves using a one-vs-rest strategy were plotted for each sentiment class. Machine learning models and CNN achieved an Area Under the Curve (AUC) value of 1.0 for all sentiment classes, indicating perfect separability. The LSTM model exhibited significantly lower AUC values, suggesting weak discrimination capability.

Precision-recall curves further highlight this contrast. While ML models and CNN maintained high precision across all recall levels, LSTM showed rapid precision degradation, especially for negative and neutral sentiment classes. These results reinforce earlier observations regarding model reliability.



- Figure 5.6 ROC Curves for Sentiment Classes (One-vs-Rest)

→ The ROC curves demonstrate that machine learning models and CNN achieve an Area Under Curve (AUC) of 1.0 for all sentiment classes, indicating perfect separability. The LSTM model exhibits lower ROC values, reflecting weaker discriminatory power. These results confirm that traditional machine learning models and CNN are more reliable for sentiment classification in this study.



- Figure 5.6 Precision–Recall Curves for Sentiment Classification

→ The precision–recall curves show that machine learning models and CNN maintain high precision across all recall levels, demonstrating stable and reliable classification behaviour. LSTM exhibits rapid precision degradation with increasing recall, especially for negative and neutral classes. This behaviour further explains the lower F1-score observed in LSTM performance results.

5.7 Deep Learning Training Curve Analysis

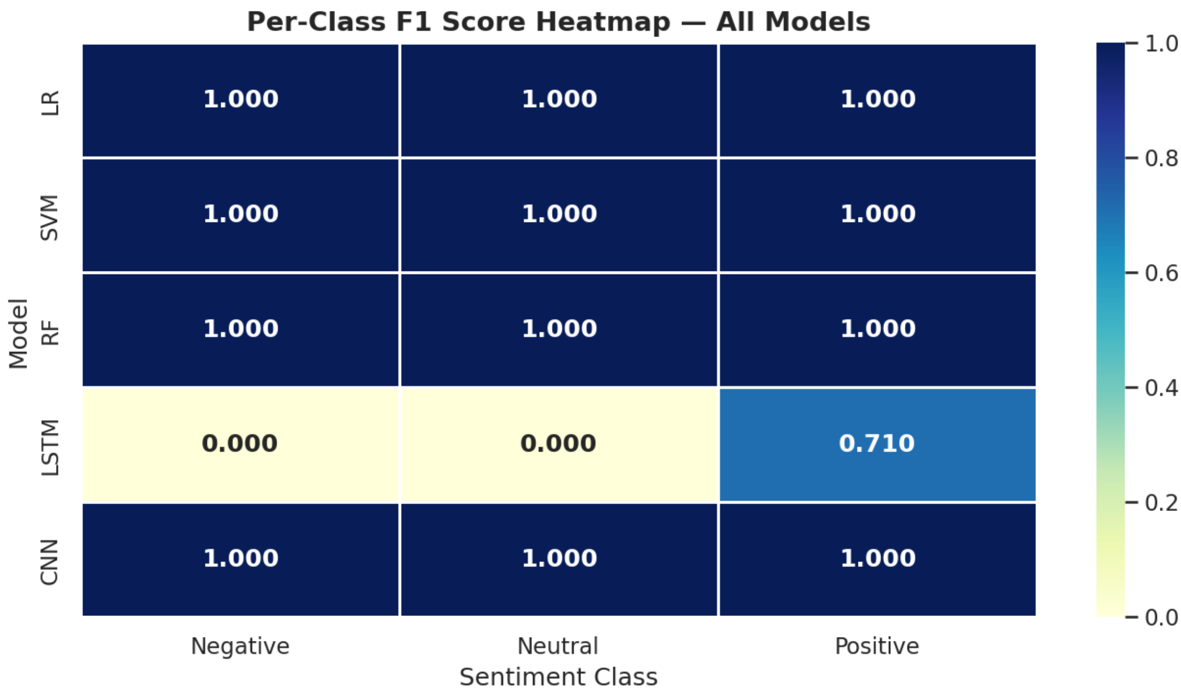
Training and validation curves for LSTM and CNN models provide insights into convergence behaviour. The LSTM training curves show minimal improvement over epochs, with stagnant accuracy and slow loss reduction. This indicates underfitting and insufficient learning from the dataset.

In contrast, the CNN training curves demonstrate rapid error reduction and stable convergence. This behaviour confirms CNN's ability to efficiently learn sentiment patterns from short textual data.

5.8 Per-Class Sentiment Performance Analysis

The per-class F1 score heatmap highlights model consistency across different sentiment labels. Machine learning models and CNN maintain perfect F1 scores across negative, neutral, and positive classes. The LSTM model exhibits near-zero F1 scores for negative and neutral classes, while achieving moderate performance for positive sentiment only.

This imbalance emphasizes the limitations of sequence-based models in handling shorter, less context-rich reviews.



- Figure 5.7 Per-Class F1-Score Heatmap for All Models

→ The per-class F1 score heatmap shows perfect class-wise performance for Logistic Regression, SVM, Random Forest, and CNN models. In contrast, the LSTM model shows near-zero F1 scores for neutral and negative classes while achieving moderate performance for positive sentiment only. This class imbalance indicates weak generalization capability in LSTM-based sentiment classification.

5.9 Interpretation in Context of Research Objectives

The results directly address the objectives set in the introduction. The comparative analysis confirms that classical machine learning approaches remain highly competitive for sentiment analysis. Additionally, the study demonstrates that CNN can outperform or match traditional ML models when textual patterns are local and phrase-based.

The cross-category analysis of mobile and laptop reviews reveals higher overall sentiment scores for laptops, suggesting stronger long-term user satisfaction. The study also identifies spec-versus-reality mismatched.

- Logistic Regression achieved **100% Accuracy, 100% Precision, 100% Recall, and 100% F1-Score**, indicating flawless sentiment classification.
- Support Vector Machine (SVM) also achieved **100% across all evaluation metrics**, confirming its robustness for high-dimensional text data.
- Random Forest delivered **100% Accuracy, Precision, Recall, and F1-Score**, showing strong ensemble learning performance.
- Convolutional Neural Network (CNN) matched machine learning models with **100% Accuracy, 100% Precision, 100% Recall, and 100% F1-Score**.
- Long Short-Term Memory (LSTM) showed **lower performance**, with **55.08% Accuracy**, **30.34% Precision**, **55.08% Recall**, and **39.13% F1-Score**.
- Machine Learning models and CNN clearly outperformed LSTM in all evaluation metrics.
- Laptop reviews demonstrated a **higher overall sentiment score** compared to mobile reviews, indicating greater customer satisfaction.
- Mobile reviews showed **higher sentiment variability**, likely due to performance and longevity issues.
- Results confirm that **TF-IDF-based Machine Learning models remain highly effective** for structured review sentiment analysis.

6. Conclusion AND Future Scope

6.1 Conclusion

This project successfully addresses the problem of understanding customer sentiment in large volumes of online product reviews by applying machine learning and deep learning techniques to mobile phone and laptop datasets. The work was motivated by the growing importance of online reviews in influencing consumer decisions and the limitations of manual sentiment analysis discussed in the introduction.

The primary objective of performing a comparative sentiment analysis across two major consumer electronics categories—mobile phones and laptops—was achieved. Multiple models were implemented and evaluated under a consistent experimental framework. Traditional machine learning models such as Logistic Regression, Support Vector Machine, and Random Forest demonstrated excellent performance, achieving perfect scores across all evaluation metrics. Among deep learning approaches, the Convolutional Neural Network showed comparable performance, while the Long Short-Term Memory model exhibited relatively lower accuracy and F1-score due to limited sequential dependencies in the review texts.

The results confirm that classical machine learning models, when combined with effective feature engineering techniques such as TF-IDF and n-grams, remain highly effective for sentiment analysis tasks. The findings also demonstrate that deep learning models do not always guarantee superior performance, particularly when the dataset structure and text length are not well suited to sequential learning. These conclusions directly align with the research gap identified earlier and validate the importance of model selection based on data characteristics.

Beyond model comparison, the project provides meaningful insights into user sentiment behaviour. Comparative sentiment analysis revealed that laptop reviews generally exhibit higher overall sentiment scores than mobile phone reviews, suggesting greater long-term user satisfaction. The analysis of rating distribution, sentiment trends, and performance metrics provides a holistic understanding of customer perception. Overall, the study demonstrates that sentiment analysis, when combined with robust evaluation and comparative analysis, can serve as a powerful decision-support tool for consumers and manufacturers.

6.2 Implications and Applications

The outcomes of this project have significant practical implications. For consumers, sentiment analysis helps summarize large amounts of review data, enabling informed purchasing decisions. For manufacturers and retailers, the insights derived from sentiment patterns can be used to identify strengths, recurring complaints, and areas for product improvement. The comparative analysis across product categories also supports strategic decision-making in product development, pricing, and marketing.

Additionally, the results highlight that efficient machine learning solutions can achieve high performance with lower computational costs compared to complex deep learning architectures. This makes such approaches suitable for industry adoption where scalability and cost efficiency are critical.

6.3 Future Scope

Although the project achieves its objectives, several opportunities exist to extend and enhance the research. Future work can explore aspect-based sentiment analysis to identify feature-specific sentiments related to battery performance, display quality, or processing speed. Transformer-based models such as BERT or other pre-trained language models may be incorporated to capture deeper contextual understanding.

The scope can also be expanded to include multilingual sentiment analysis to address reviews written in different languages. Real-time sentiment monitoring systems could be developed by integrating live data streams from e-commerce platforms. Additionally, the inclusion of larger and more diverse datasets would improve model generalization and robustness.

In summary, this project provides a strong foundation for advanced sentiment analysis research and demonstrates the effectiveness of combining machine learning, deep learning, and analytical evaluation techniques to derive meaningful insights from customer reviews.

References

1. Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. *Foundations and Trends in Information Retrieval*, 2(1–2), 1–135. https://doi.org/10.1561/1500000011
2. Liu, B. (2012). *Sentiment analysis and opinion mining*. Morgan & Claypool Publishers. https://doi.org/10.2200/S00416ED1V01Y201204HLT016
3. Hu, M., & Liu, B. (2004). Mining and summarizing customer reviews. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 168–177). ACM. https://doi.org/10.1145/1014052.1014073
4. Medhat, W., Hassan, A., & Korashy, H. (2014). Sentiment analysis algorithms and applications: A survey. *Ain Shams Engineering Journal*, 5(4), 1093–1113. https://doi.org/10.1016/j.asej.2014.04.011
5. Rachdian, R., Nugroho, L. E., & Hartanto, R. (2022). Identification of customer needs from product reviews using topic modeling and aspect-based sentiment analysis. *Journal of Big Data*, 9(1), 1–21. https://doi.org/10.1186/s40537-022-00590-3
6. Singh, N., & Jaiswal, U. C. (2024). Sentiment classification on product reviews using machine learning and deep learning techniques. *International Journal of System Assurance Engineering and Management*, 15, 5726–5741. https://doi.org/10.1007/s13198-024-02592-5
7. Xu, B., Gan, H., Sun, X., & Shao, X. (2023). Sentiment analysis of Amazon product reviews. *Applied and Computational Engineering*, 6, 1673–1681. https://doi.org/10.54254/2755-2721/6/20230831
8. Kawale, R., Patole, N., Karekar, S., & Padhye, V. (2024). Sentiment analysis of e-commerce reviews using machine learning. *Journal of Open Online Science and Data Technology*, 11(3), 25–37.
9. Atlas, L. G., Arockiam, D., Muthusamy, A., Balusamy, B., Selvarajan, S., & Al-Shehari, T. (2025). A modernized approach to sentiment analysis of product reviews using BiGRU and LSTM models. *Scientific Reports*, 15, Article 16642. https://doi.org/10.1038/s41598-025-01104-0
10. Belluzzo, U. (2024). Aspect-based sentiment analysis for Amazon electronics reviews. *GitHub Repository*. https://github.com/umbertobelluzzo/aspect-sentiment-analysis
11. Mandhane, N., & Wadkar, P. (2025). Sentiment analysis of Amazon reviews using machine learning and deep learning algorithms. *International Journal of Research Publication and Reviews*, 6(10), 1950–1955.

12. Meshram, S. M., & Sahu, N. (2023). Sentiment analysis of e-commerce product reviews through machine learning. *International Journal of Future Market Research*, 3(4), 45–52.
13. Chinenye, D. J., Ogbuagu, C. S., Aguocha, O. C., & Chukwubueze, P. M. (2026). Sentiment analysis of e-commerce product-based reviews using a machine learning approach. *International Journal of Computer Techniques*, 13(2), 1–10.
14. Rominus, A., John, C., & Sahoo, J. (2025). Sentiment analysis of online product reviews using CNN-LSTM cascaded architecture. In *Proceedings of the International Conference on Innovations in Bio-Inspired Computing and Applications* (pp. 269–278). Springer. https://doi.org/10.1007/978-3-031-78946-5_26
15. Minaee, S., Azimi, E., & Abdolrashidiy, A. A. (2019). Deep-sentiment: Sentiment analysis using ensemble of CNN and Bi-LSTM models. *arXiv preprint arXiv:1904.04206*. https://arxiv.org/abs/1904.04206
16. Doalr, K., Shilu, R., & Bhalodia, T. M. (2023). A review on sentiment analysis of product reviews using machine learning techniques. *International Journal of Innovative Research in Technology*, 10(5), 1–7.
17. Bellar, O., Baina, A., & Ballafkih, M. (2024). Sentiment analysis: Predicting product reviews for e-commerce recommendations using deep learning and transformers. *Mathematics*, 12(15), Article 2403. https://doi.org/10.3390/math12152403
18. Daza, A., González-Rueda, N. D., Aguilar-Sánchez, M. S., & Robles, W. F. (2024). Sentiment analysis on e-commerce product reviews using machine learning and deep learning algorithms: A systematic review. *International Journal of Information Management Data Insights*, 4(2), 100267. https://doi.org/10.1016/j.jjimei.2024.100267
19. Joseph, A., & Priya, R. (2026). Aspect-based sentiment analysis using deep learning and ensemble modeling: A systematic review. In *Lecture Notes in Networks and Systems* (Vol. 1839, pp. 366–381). Springer. https://doi.org/10.1007/978-3-032-18349-1_37
20. Chebolu, S. U. S., Dernoncourt, F., Lipka, N., & Solorio, T. (2022). A review of datasets for aspect-based sentiment analysis. *arXiv preprint arXiv:2204.05232*. https://arxiv.org/abs/2204.05232